

JavaScript Key Learnings

UI Assessment Prep — DOM, localStorage & JSON

1. HTML Input Elements

Different input types serve different purposes. Every element should have a unique **id** so JavaScript can find it.

```
<input type="text" id="username"> <!-- text box -->
<input type="color" id="themeColor"> <!-- color picker -->
<button onclick="saveSettings()">Save</button>
<h2 id="heading">Welcome</h2>
```

■ **Key rule:** The **id** is a unique name tag — it must not repeat on the page.

2. Selecting Elements — getElementById

Use **document.getElementById()** to grab an element from the page into a variable.

```
let inputEl = document.getElementById("username");
// inputEl now holds the actual DOM element (the input box)
```

3. Reading User Input — .value

Once you have the element, read what the user typed using **.value**.

```
let inputEl = document.getElementById("username");
let name = inputEl.value; // e.g. "John"

let colorEl = document.getElementById("themeColor");
let color = colorEl.value; // e.g. "#4a90d9"
```

■ **Mental model:** Element = the box. **.value** = what's inside the box.

4. localStorage — Browser Storage

localStorage lets you save data in the browser that **survives page refresh**.

Action	Code
Save	localStorage.setItem("key", "value")
Read	localStorage.getItem("key")
Delete	localStorage.removeItem("key")

■■ **Important:** localStorage only stores **strings** — not objects. Use JSON to convert.

5. JSON — Stringify & Parse

Since localStorage only takes strings, you must convert your object to a string first.

```
const obj = { name: "John", color: "#4a90d9" };

// Pack (object → string)
const str = JSON.stringify(obj);
// str = '{"name":"John","color":"#4a90d9"}'

// Unpack (string → object)
const back = JSON.parse(str);
// back.name → "John"
// back.color → "#4a90d9"
```

■ **Flow:** *stringify* → *setItem* | *getItem* → *parse*

6. DOM Manipulation

After reading saved data, update the page visually.

```
// Change page background
document.body.style.backgroundColor = data.color;

// Change text inside an element
document.getElementById("heading").textContent = "Welcome back, " + data.name;
```

7. Event Handling — addEventListener

Attach a function to a button click using **addEventListener**.

```
const btn = document.getElementById("btn");

btn.addEventListener("click", function() {
  // runs every time the button is clicked
});
```

■■ Common Bug: Overwriting Element References

The most common mistake in this question:

■ **Wrong** — destroys the element reference:

```
let inpVal = document.getElementById("username");
inpVal = inpVal.value; // inpVal is now a string – breaks on 2nd click!
```

■ **Correct** — keep element, read value into a new variable:

```
let inpEl = document.getElementById("username");
let name = inpEl.value; // inpEl stays as the element – works every time!
```

■■ Full Correct Flow

1. Get elements with `getElementById` (once, outside the function)
2. On button click:
 - a. Read `.value` into NEW variables
 - b. Bundle into an object { `inp: name, col: color` }

- c. `JSON.stringify(obj)` → pass into `localStorage.setItem`
- d. `localStorage.getItem` → `JSON.parse` → data object
- e. `document.body.style.backgroundColor = data.col`
- f. `h2.textContent = "Welcome back, " + data.inp`